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Risk Budgeting: Risk-Based Approaches to Asset Allocation

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Course Objectives

- Appreciate the recent developments in portfolio construction, including moving from asset classes to risk classes and their correlations
- Understand the role of value at risk and expected shortfall in portfolio management
- Applying the principles of Risk Budgeting in practice in the context of fixed income portfolios
- Implement risk budgeting approaches in Octave, Matlab, Python

Course Content

- Tab 1: Portfolio Construction and Measurement
- Tab 2: Value at Risk in Portfolio Management
- Tab 3: Risk Budgeting – Theory
- Tab 4: Risk Budgeting – Practice

Key References

- [1] Neil D. Pearson, *Risk Budgeting: Portfolio Problem Solving with Value-at-Risk*, Wiley, 2002
- [2] Bernd Scherer, *Portfolio Construction and Risk Budgeting* (5th Edition), Risk Books, 2015
- [3] Benjamin Bruder and Thierry Roncalli, *Managing Risk Exposures using the Risk Budgeting Approach*, **Lyxor Asset Management**, Paris, 2012. [<http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.642.6710&rep=rep1&type=pdf>]
- [4] Thomas M. Idzorek, *Risk Budgeting—Where Do You Spend Your Risk?*, **Morningstar**, 2008. [<http://gladmainnew.morningstar.com/clientcomm/RiskBudgeting.pdf>]
- [5] Patricia Vázquez Hinojosa, *Forecasting the Ex-Ante Tracking Error for Global Fixed Income Portfolios: emphasis on the estimation of the variance-covariance matrix*, Trabajo de investigación 013/016, Universidad Complutense de Madrid, 2013. [<https://www.uv.es/qfinan/TFM-numeros/13-016>]
- [6] Hakan Kaya, Wai Lee, and Yi Wan, "Risk Budgeting with Asset Class and Risk Class Approaches", *Journal of Investing*, Vol. 21, No. 1 (Spring 2012): 109-115. [<http://www.cfapubs.org/doi/full/10.2469/dig.v42.n3.36>]
- [7] Phil Joubert and Stephen Langdell, "Fixing a broken correlation matrix", The Numerical Algorithms Group, Oxford, UK. [<https://www.nag.co.uk/IndustryArticles/fixing-a-broken-correlation-matrix.pdf>]
- [8] GNU Octave, Scientific Programming Language, <https://www.gnu.org/software/octave/>.

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Risk Budgeting

*The process of decomposing the aggregate risk of a portfolio into its constituents, using these risk measures to allocate assets, setting limits in terms of these measures, and then using the limits to monitor the asset allocations and portfolio managers is known as risk allocation or risk budgeting.**

*Extract from Neil D. Pearson, Risk budgeting: Portfolio Problem Solving with Value-at-Risk, Wiley 2002, page 3

Risk Budgeting

Narrowly defined, **risk budgeting** is a process of measuring and decomposing risk, using the measures in asset-allocation decisions, assigning portfolio managers risk budgets defined in terms of these measures, and using these risk budgets in monitoring the asset allocations and portfolio managers.

A prerequisite for risk budgeting is risk decomposition, which involves:

- Identifying the various sources of risk, or risk factors, such as equity returns, interest rates, and exchange rates;
- Measuring each factor's, manager's, and asset class's contribution to the total risk;
- Comparing the ex post realised outcomes to the ex-ante risk; and
- Identifying the risks that were taken intentionally, and those taken inadvertently.

Risk Budgeting

This risk decomposition allows a plan sponsor to have a better understanding of the risks being assumed and how they have changed, and to have more informed conversations with the portfolio managers. In the event that there are problems, it allows the sponsor to identify unwanted risks and managers who deviate from their historical styles before any negative outcomes occur.

Risk Budgeting

If this risk decomposition is combined with an explicit set of risk allocations to factors, managers, or asset classes, it is called risk allocation or risk budgeting. The risk budgeting process itself consists of:

- Setting limits, or risk budgets, on the quantity of risk due to each asset class, manager, or factor;
- Establishing asset allocations based on the risk budgets;
- Comparing the risk budgets to the measures of the risk due to each factor on an ongoing basis; and
- Adjusting the asset allocations to keep the risks within the budgeted limits.

Risk Budgeting

Risk decomposition is crucial to risk budgeting, because the aggregate value at risk of the pension plan or other organisation is far removed from the portfolio managers. At the risk of stating the obvious, the portfolio managers have control only over their own portfolios. For them, meaningful risk budgets are expressed in terms of their contributions to portfolio risk.

However, risk budgeting is more than a list of steps or procedures. Defined more broadly, risk budgeting is a way of thinking about investment and portfolio management. For this reason, to find a definition that attracts broad agreement is difficult, and perhaps impossible. The world view that underlies risk budgeting takes for granted reliance upon probabilistic or statistical measures of risk and the use of modern risk – and portfolio management tools to manage risk. Thinking about the asset-allocation problem in terms of risk allocations rather than traditional asset allocations is a natural outgrowth of this world view.

Risk Budgeting

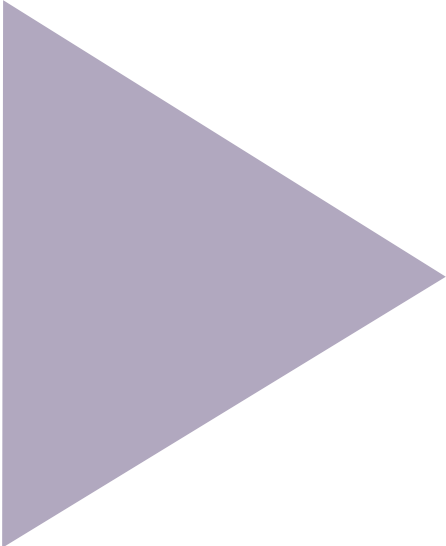
From a logical perspective, there is no special relation between value at risk and risk budgeting. Risk budgeting requires a measure of portfolio risk, and value at risk is one candidate. It is a natural candidate, in that:

- i. It is a measure of downside risk, and thus useful when the distribution of portfolio returns is asymmetric; and
- ii. When returns are normally distributed, it is equivalent to a forward-looking estimate of portfolio standard deviation.

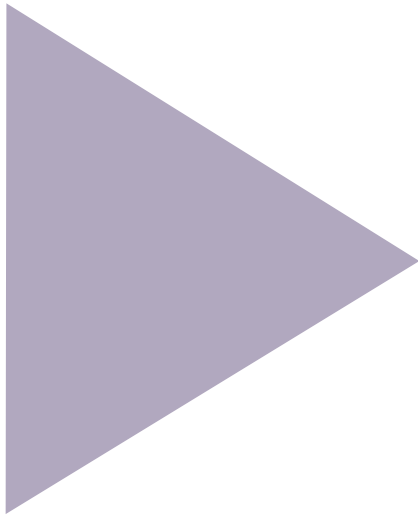
However, the risk budgeting process could be implemented using any of a number of risk measures. For example, it could be implemented using either a forward-looking estimate of portfolio standard deviation or a scenario-based measure. In fact, it is widely recommended that value at risk measures be used in combination with stress testing (procedures to estimate the losses that might be incurred in extreme or 'stress' scenarios).

Risk Budgeting

In practice, however, value at risk and risk budgeting are intimately related. Because risk budgeting involves the quantification, aggregation, and decomposition of risk, the availability of a well-recognised aggregate measure of portfolio risk is a prerequisite for its use and acceptance. In this sense, risk budgeting is an outgrowth of value at risk. But for the popularity and widespread acceptance of value at risk, you would likely not be hearing and reading about risk budgeting today. Nonetheless, value at risk has some well known limitations, and it may be that some other risk measure eventually supplants value at risk in the risk budgeting process.



Tab 1: Portfolio Construction and Measurement



Risk Return Diagrams

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Mean-Standard Deviation Diagrams

The random rates of return of assets can be represented on two-dimensional diagram.



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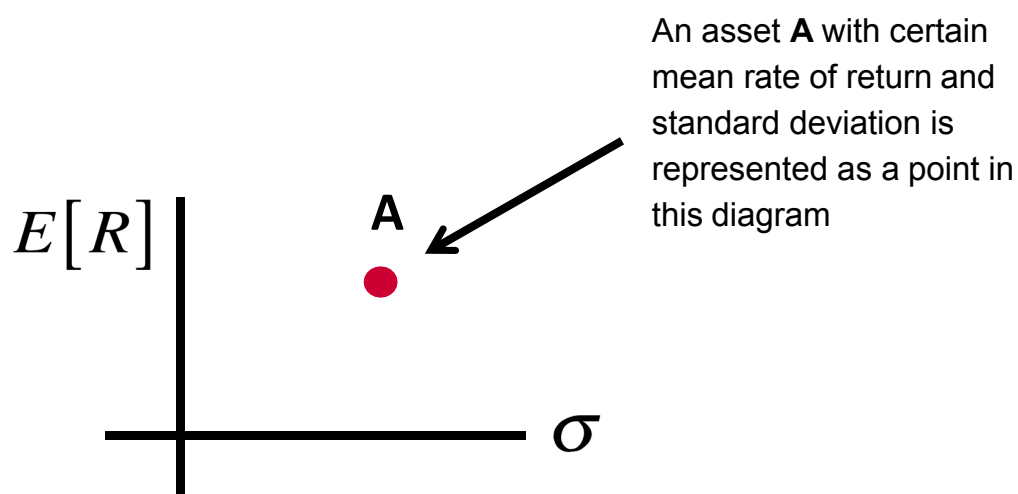
Mean-Standard Deviation Diagrams

The horizontal axis is used for the standard deviation, and the vertical axis is used for the mean.

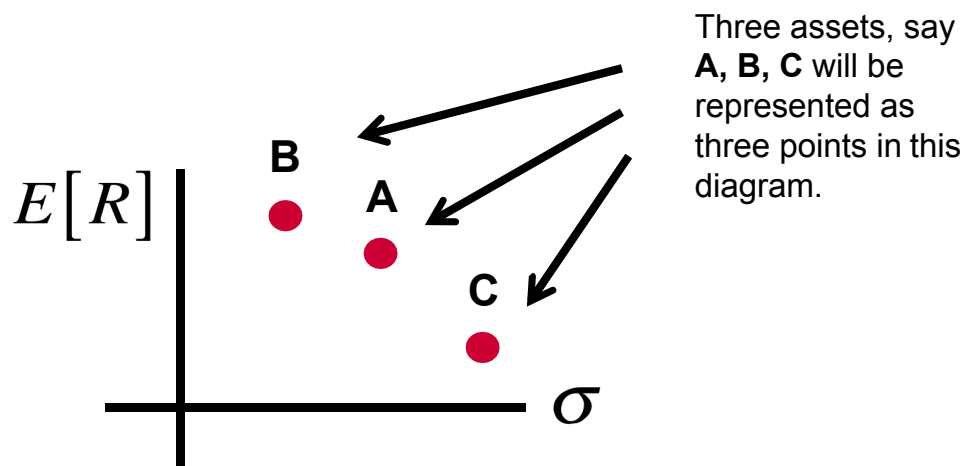
This diagram is called a mean-standard deviation diagram or the risk-return diagram.



Mean-Standard Deviation Diagrams



Mean-Standard Deviation Diagrams

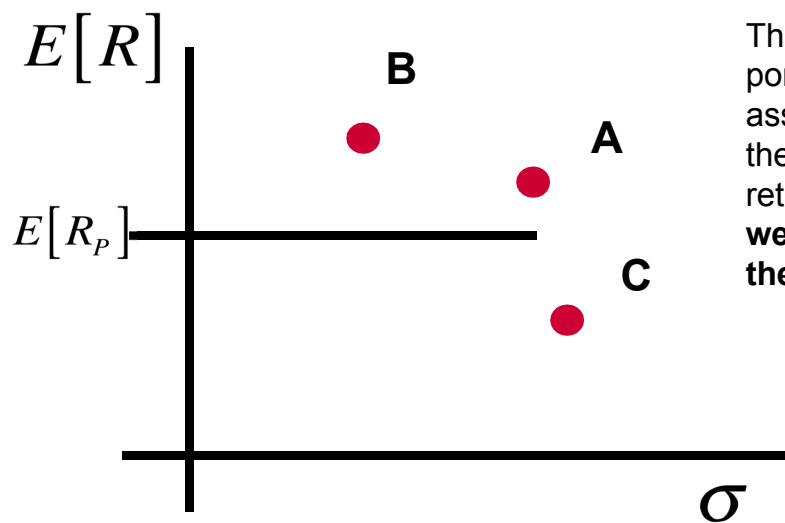


Mean-Standard Deviation Diagrams

$$E[R_p] = w_1 E[R_1] + w_2 E[R_2] + \dots + w_N E[R_N]$$

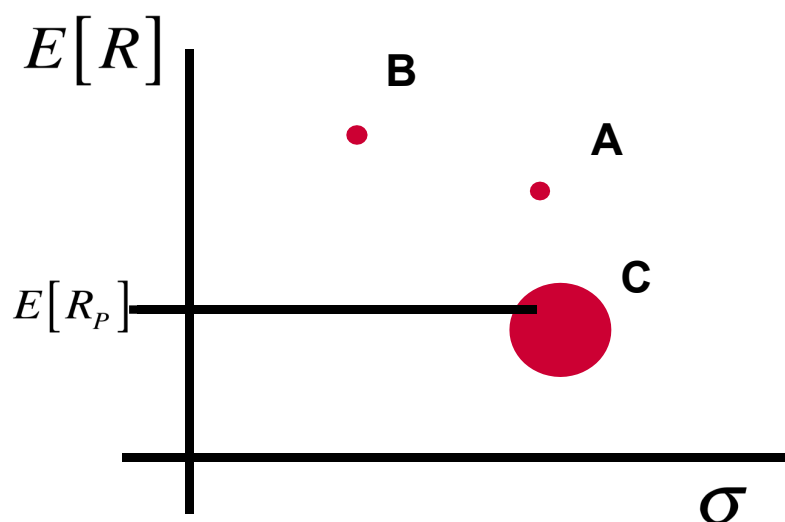
$$E[R_p] = \sum_{i=1}^N w_i E[R_i]$$

Mean-Standard Deviation Diagrams



The mean return of a portfolio of three assets, **A**, **B**, **C** is not the mean of the returns but the **weighted-mean of the returns**.

Mean-Standard Deviation Diagrams



So we must consider the **weight** of each asset with respect to the portfolio.

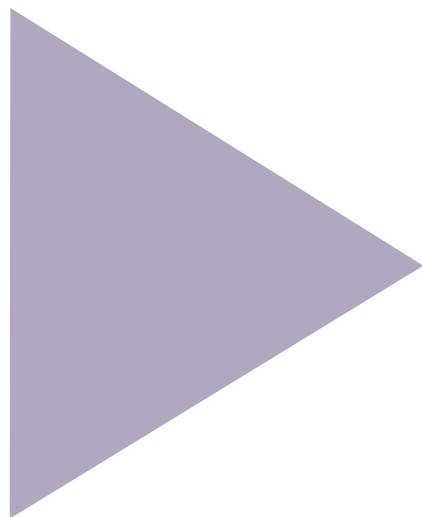
If **WA=10%**,
WB=10%,
WC=80%, then the mean return of the portfolio would look like...

Portfolio Variance

$$\sigma_P^2 = E\left[(R - \bar{R})^2\right]$$

$$\sigma_P^2 = E\left[\left(\sum_{i=1}^N w_i R_i - \sum_{i=1}^N w_i \bar{R}_i\right)^2\right]$$

$$\sigma_P^2 = \sum_{i,j=1}^N w_i w_j \sigma_{i,j}$$



Modern Portfolio Theory (MPT)

Modern Portfolio Theory

Two Asset Portfolio

Consider a portfolio composed of two assets, A and B.

Their respective returns and volatilities are:

$$A : (R_A, \sigma_A)$$

$$B : (R_B, \sigma_B)$$

And their weights:

$$W_A = 1 - \alpha$$

$$W_B = \alpha$$

Modern Portfolio Theory

Two Asset Portfolio

The correlation between the returns of the two assets **A** and **B**, can be approximated with the Pearson correlation coefficient between the two time series:

$$\rho = [-1, +1]$$

Modern Portfolio Theory

Two Asset Portfolio

The portfolio return R_p based on the returns of the two assets **A** and **B** is then

$$R_p = W_A R_A + W_B R_B$$
$$R_p = (1 - \alpha) R_A + \alpha R_B$$

Modern Portfolio Theory

Two Asset Portfolio

The portfolio volatility is based on the covariance between the two assets **A** and **B**

$$\sigma_p = \sqrt{(1 - \alpha)^2 \sigma_A^2 + 2\alpha(1 - \alpha)\sigma_{AB} + \alpha^2 \sigma_B^2}$$

Using the definition

$$\rho = \frac{\sigma_{AB}}{\sigma_A \sigma_B} \rightarrow \sigma_{AB} = \sigma_A \sigma_B \rho$$

It can be written as

$$\sigma_p = \sqrt{(1 - \alpha)^2 \sigma_A^2 + 2\alpha(1 - \alpha)\sigma_A \sigma_B \rho + \alpha^2 \sigma_B^2}$$

Modern Portfolio Theory

Two Asset Portfolio

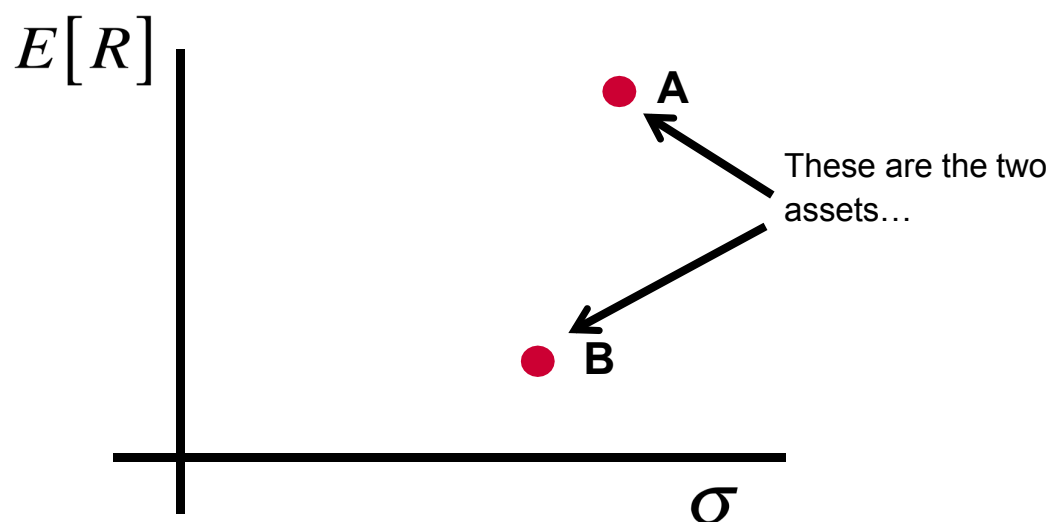
We can then describe the return and volatility of the portfolio by the master equations:

$$R_P = (1 - \alpha)R_A + \alpha R_B$$

$$\sigma_P = \sqrt{(1 - \alpha)^2 \sigma_A^2 + 2\alpha(1 - \alpha)\sigma_A \sigma_B \rho + \alpha^2 \sigma_B^2}$$

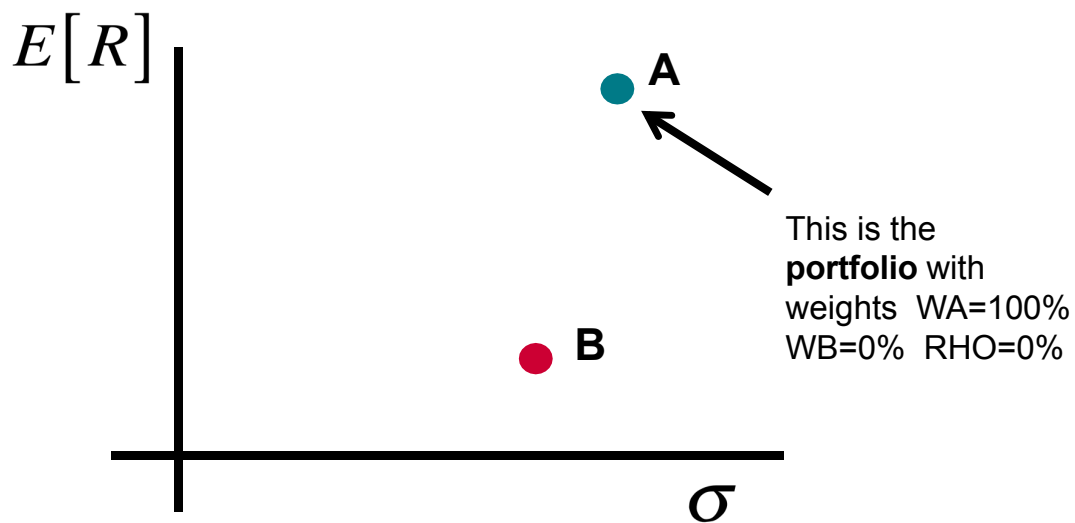
Modern Portfolio Theory

Two Asset Portfolio



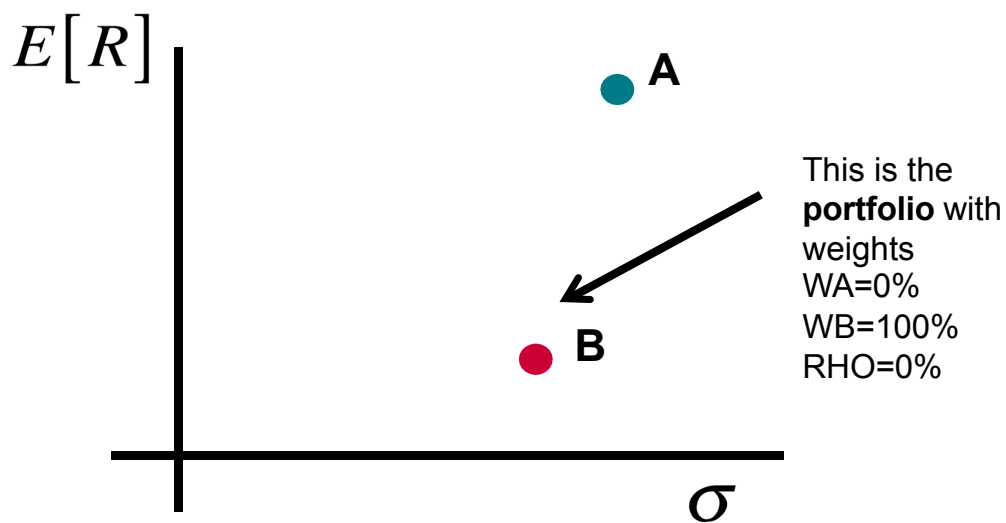
Modern Portfolio Theory

Two Asset Portfolio



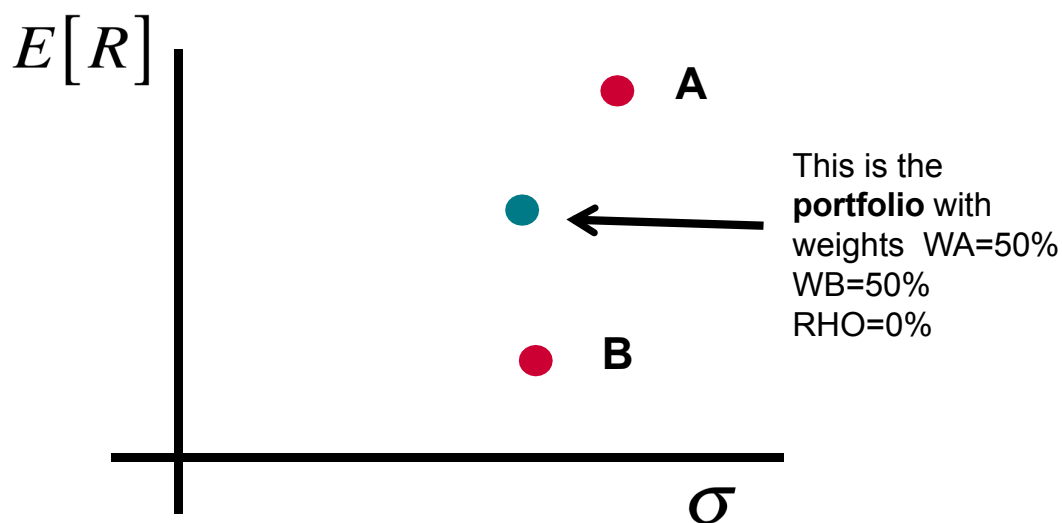
Modern Portfolio Theory

Two Asset Portfolio



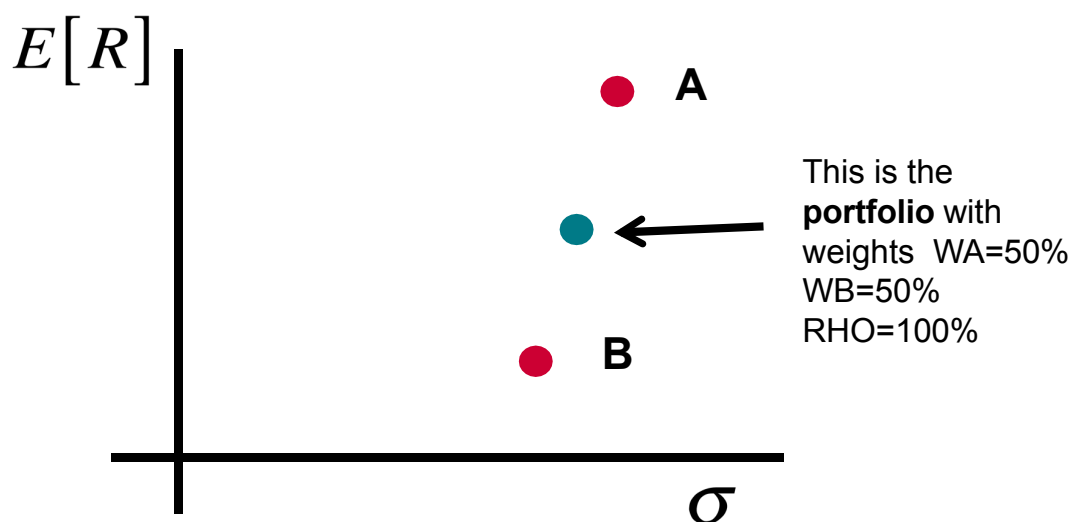
Modern Portfolio Theory

Two Asset Portfolio



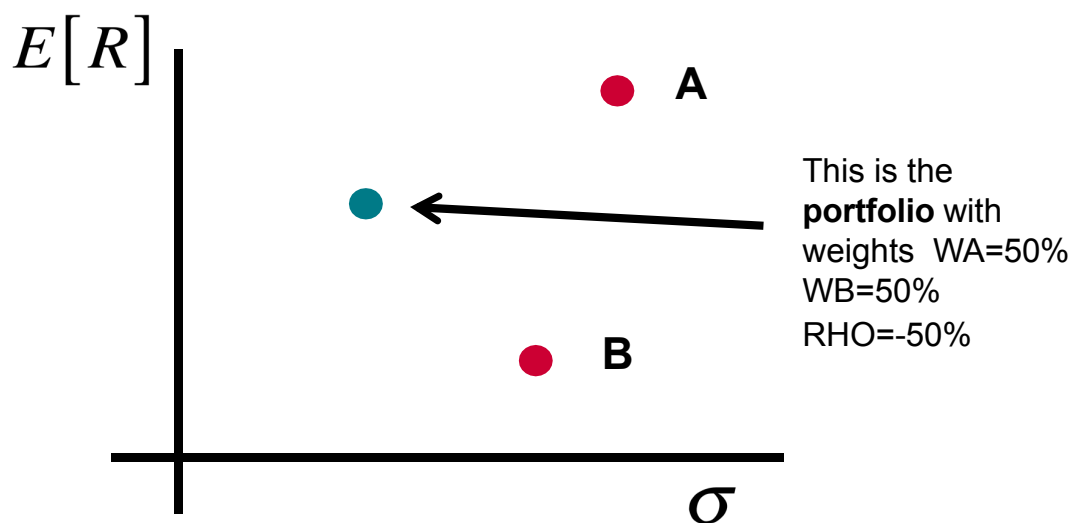
Modern Portfolio Theory

Two Asset Portfolio



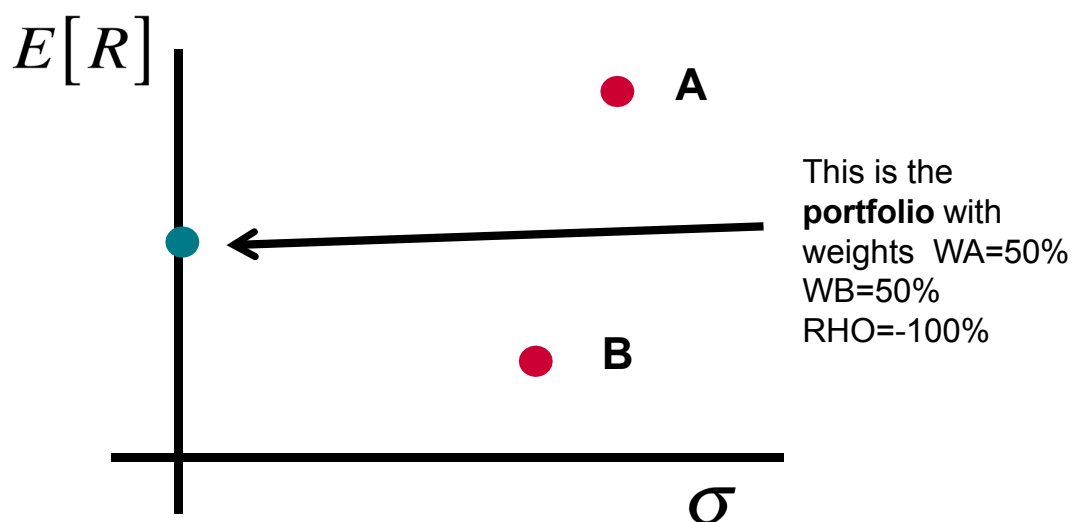
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Two Asset Portfolio



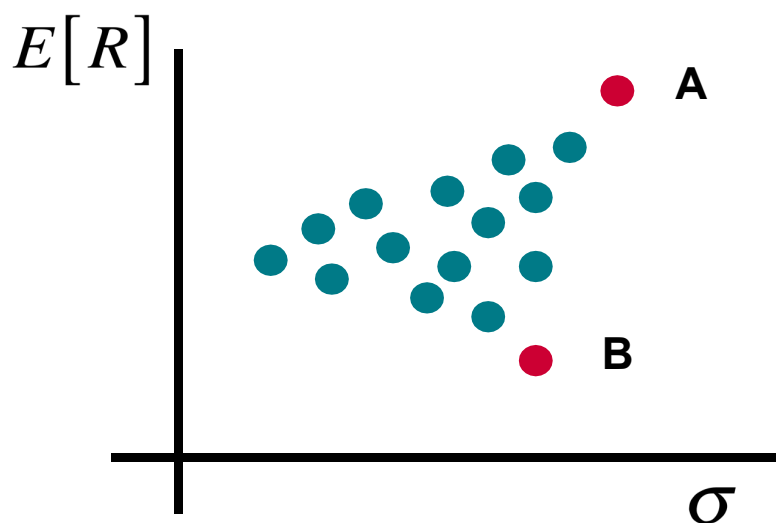
Modern Portfolio Theory

Two Asset Portfolio



Modern Portfolio Theory

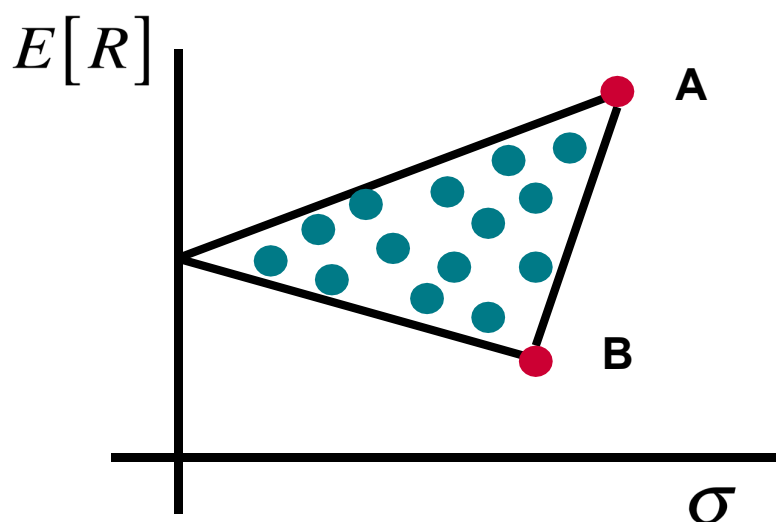
Two Asset Portfolio



These are the **portfolios** than can be obtained by changing the weights and the correlation:
WA
WB
RHO

Modern Portfolio Theory

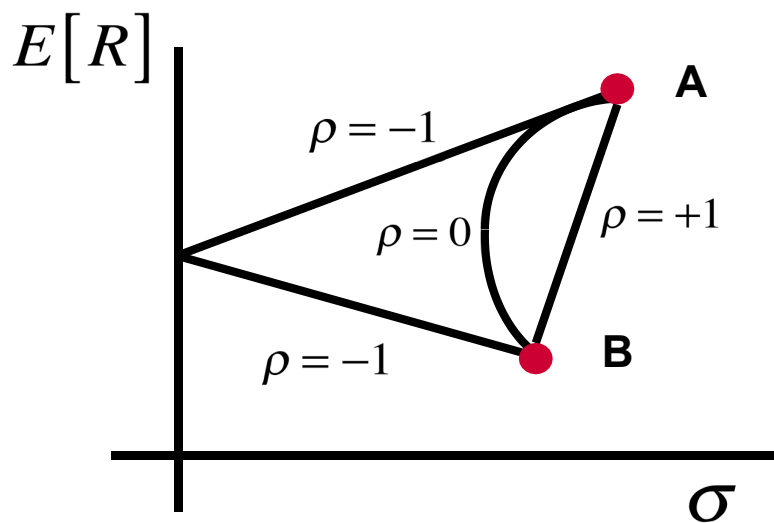
Two Asset Portfolio



It can be shown that the possible (**attainable region**) of all the possible portfolios is...

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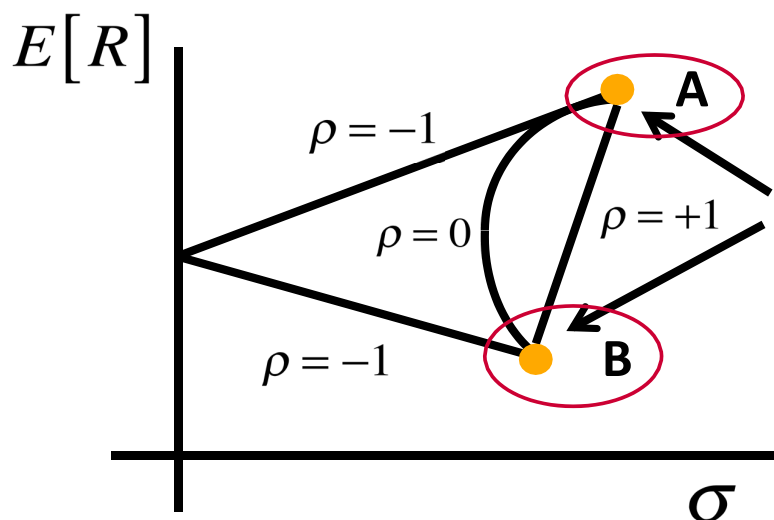
Two Asset Portfolio



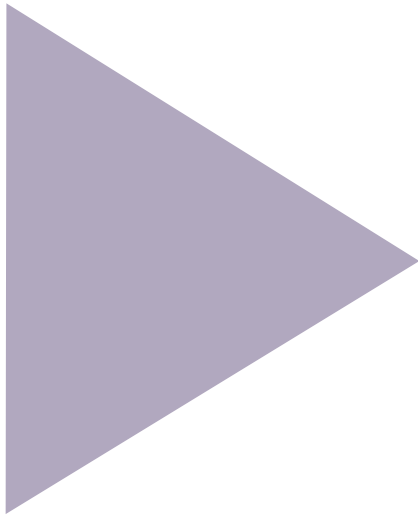
And in terms of the correlation...

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Two Asset Portfolio



Note that the location of these remain fixed... we can call them portfolio anchors...

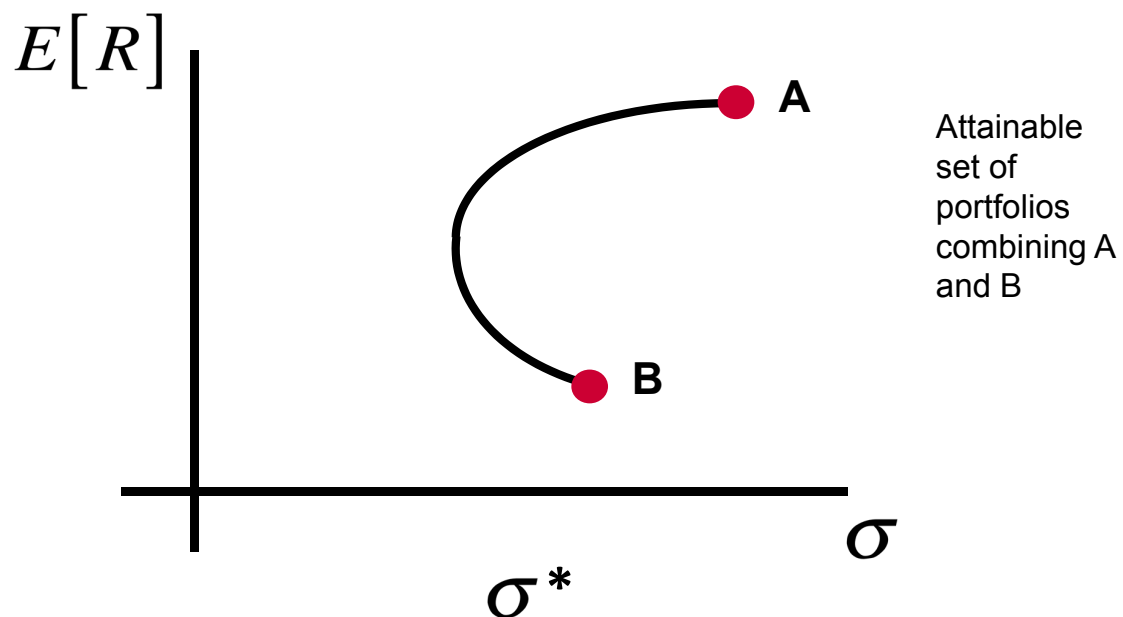


Efficient Frontier

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Modern Portfolio Theory

Efficient Frontier

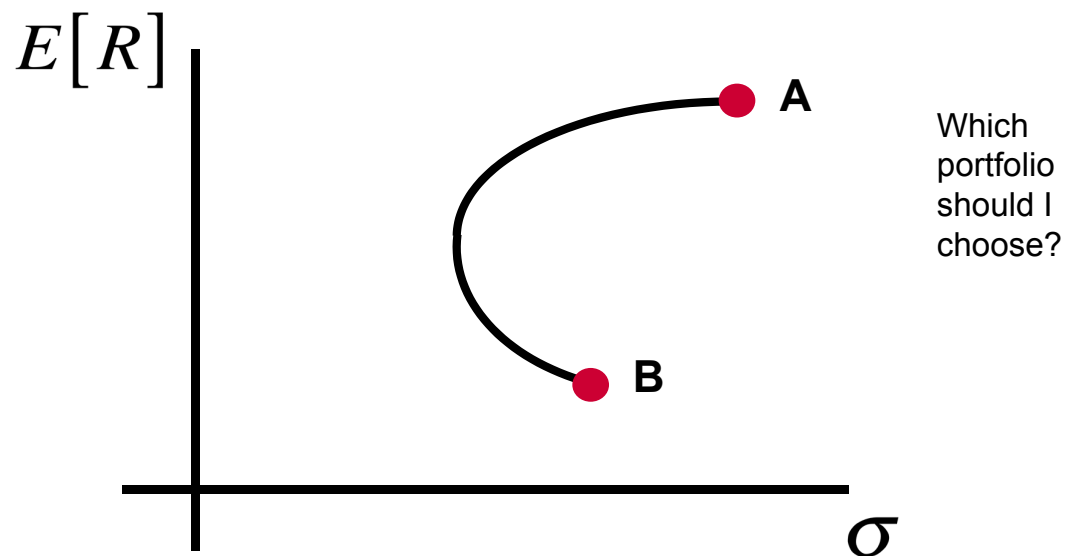


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Two Asset Portfolio



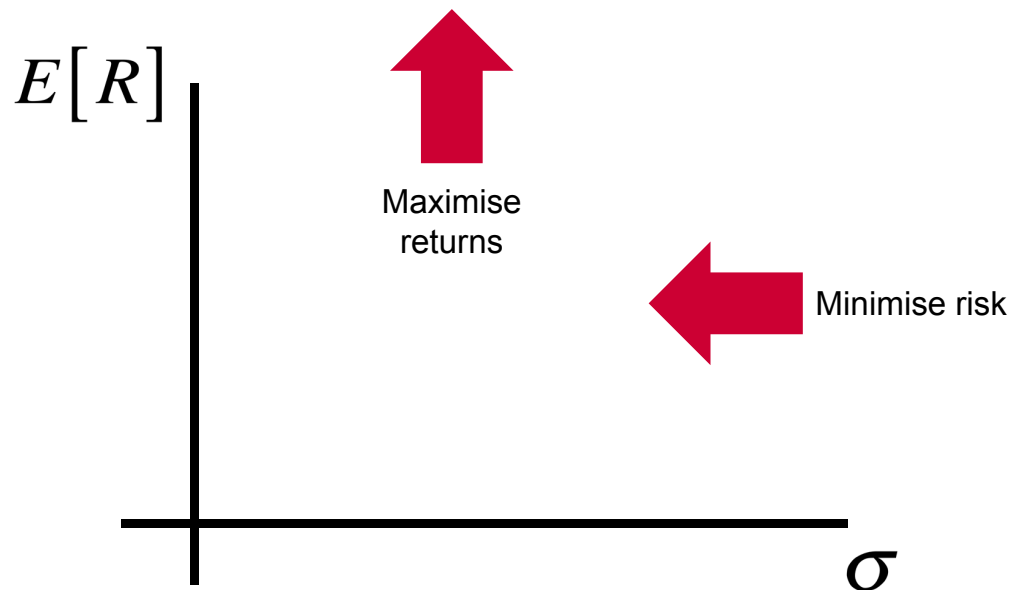
Modern Portfolio Theory

Two Asset Portfolio



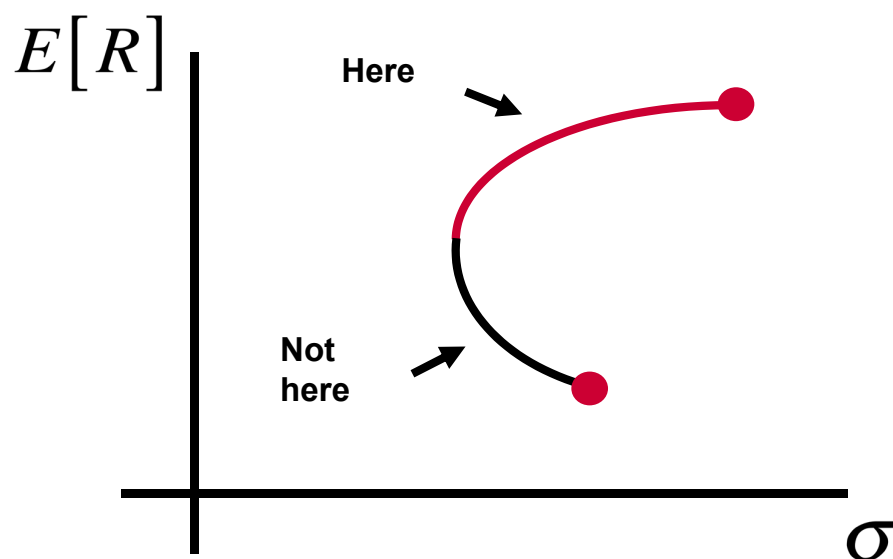
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Two Asset Portfolio



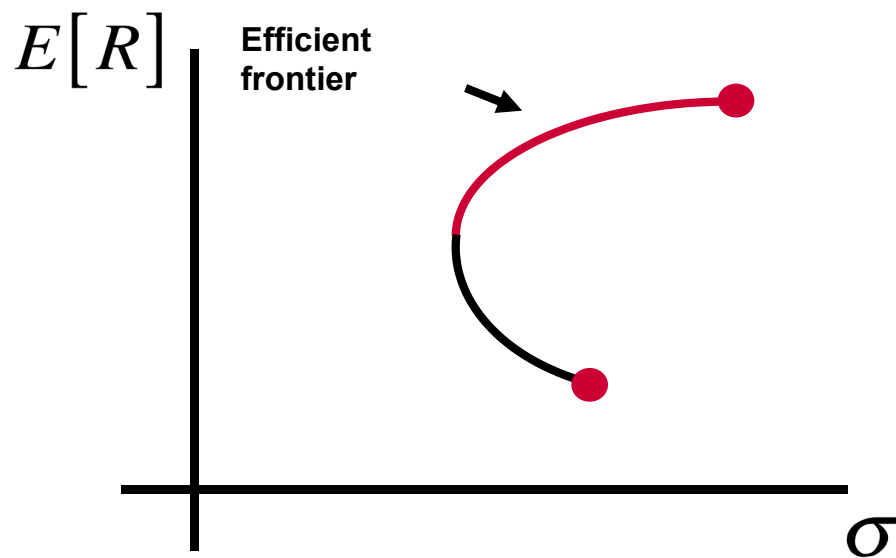
Modern Portfolio Theory

Two Asset Portfolio



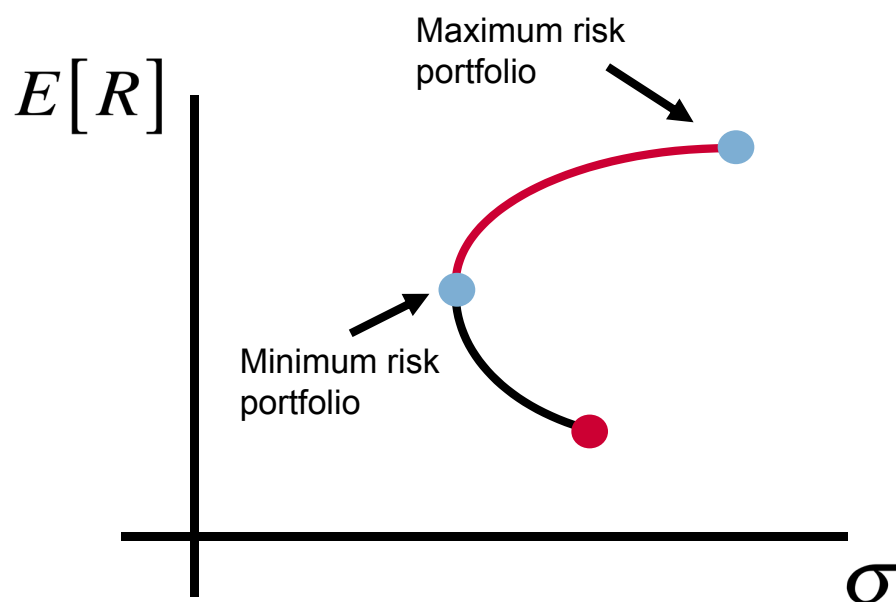
Modern Portfolio Theory

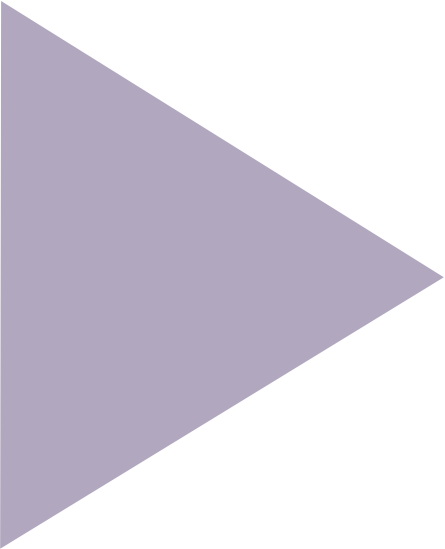
Two Asset Portfolio



Modern Portfolio Theory

Two Asset Portfolio





Modern Portfolio Theory

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Harry Markowitz



Harry Max Markowitz (born 24 August 1927) is an American economist, and a recipient of the 1989 John von Neumann Theory Prize and the 1990 Nobel Memorial Prize in Economic Sciences.

<https://www.thefamouspeople.com/profiles/images/harry-markowitz-1.jpg>

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PORTFOLIO SELECTION*

HARRY MARKOWITZ
The Rand Corporation

THE PROCESS OF SELECTING a portfolio may be divided into two stages. The first stage starts with observation and experience and ends with beliefs about the future performances of available securities. The second stage starts with the relevant beliefs about future performances and ends with the choice of portfolio. This paper is concerned with the second stage. We first consider the rule that the investor does (or should) maximize discounted expected, or anticipated, returns. This rule is rejected both as a hypothesis to explain, and as a maximum to guide investment behavior. We next consider the rule that the investor does (or should) consider expected return a desirable thing *and* variance of return an undesirable thing. This rule has many sound points, both as a maxim for, and hypothesis about, investment behavior. We illustrate geometrically relations between beliefs and choice of portfolio according to the “expected returns—variance of returns” rule.

One type of rule concerning choice of portfolio is that the investor does (or should) maximize the discounted (or capitalized) value of future returns.¹ Since the future is not known with certainty, it must be “expected” or “anticipated” returns which we discount. Variations of this type of rule can be suggested. Following Hicks, we could let “anticipated” returns include an allowance for risk.² Or, we could let the rate at which we capitalize the returns from particular securities vary with risk.

The hypothesis (or maxim) that the investor does (or should) maximize discounted return must be rejected. If we ignore market imperfections the foregoing rule never implies that there is a diversified portfolio which is preferable to all non-diversified portfolios. Diversification is both observed and sensible; a rule of behavior which does not imply the superiority of diversification must be rejected both as a hypothesis and as a maxim.

Modern Portfolio Theory

“The investor does (or should) consider expected return a desirable thing and variance of return an undesirable thing.”

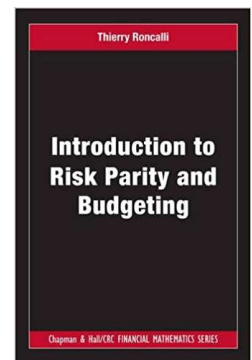
Harry Markowitz (1952)

Indeed, Markowitz showed that an efficient portfolio is the portfolio that maximises the expected return for a given level of risk (corresponding to the variance of portfolio return).

Modern Portfolio Theory

We consider a universe of n assets. Let $x = (x_1, \dots, x_n)$ be the vector of weights in the portfolio. We assume that the portfolio is fully invested meaning that $\sum_{i=1}^n x_i = 1^\top x = 1$. We denote $R = (R_1, \dots, R_n)$ the vector of asset returns where R_i is the return of asset i . The return of the portfolio is then equal to $R(x) = \sum_{i=1}^n x_i R_i$. In a matrix form, we also obtain $R(x) = x^\top R$. Let $\mu = \mathbb{E}[R]$ and $\Sigma = \mathbb{E}[(R - \mu)(R - \mu)^\top]$ be the vector of expected returns and the covariance matrix of asset returns. The expected return of the portfolio is:

$$\mu(x) = \mathbb{E}[R(x)] = \mathbb{E}[x^\top R] = x^\top \mathbb{E}[R] = x^\top \mu$$



*Extract from Thierry Roncalli, Introduction to Risk Parity and Budgeting, CRC Press, 2014

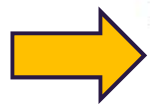
Modern Portfolio Theory

whereas its variance is equal to:

$$\begin{aligned}\sigma^2(x) &= \mathbb{E}[(R(x) - \mu(x))(R(x) - \mu(x))^\top] \\ &= \mathbb{E}[(x^\top R - x^\top \mu)(x^\top R - x^\top \mu)^\top] \\ &= \mathbb{E}[x^\top (R - \mu)(R - \mu)^\top x] \\ &= x^\top \mathbb{E}[(R - \mu)(R - \mu)^\top] x \\ &= x^\top \Sigma x\end{aligned}$$

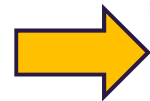
Modern Portfolio Theory

We can then formulate the investor's financial problem as follows:



1. Maximizing the expected return of the portfolio under a volatility constraint (σ -problem):

$$\max \mu(x) \quad \text{u.c.} \quad \sigma(x) \leq \sigma^* \quad (1.1)$$



2. Or minimizing the volatility of the portfolio under a return constraint (μ -problem):

$$\min \sigma(x) \quad \text{u.c.} \quad \mu(x) \geq \mu^* \quad (1.2)$$

Example 1

Minimum variance portfolio



Modern Portfolio Theory

Example 1 We consider four assets. Their expected returns are equal to 5%, 6%, 8% and 6% while their volatilities are equal to 15%, 20%, 25% and 30%. The correlation matrix of asset returns is given by the following matrix:

$$C = \begin{pmatrix} 1.00 & & & \\ 0.10 & 1.00 & & \\ 0.40 & 0.70 & 1.00 & \\ 0.50 & 0.40 & 0.80 & 1.00 \end{pmatrix}$$

Modern Portfolio Theory

By considering all the portfolios belonging to the simplex set defined by $\{x \in [0, 1]^n : 1^\top x = 1\}$, we can compute the expected return and volatility bounds of the portfolios: $\mu^- \leq \mu(x) \leq \mu^+$ and $\sigma^- \leq \sigma(x) \leq \sigma^+$. There is also a solution to the first problem if $\sigma^* \geq \sigma^-$. The second problem has a solution if $\mu^* \leq \mu^+$. If these two conditions are verified, the inequality constraints becomes $\sigma(x) = \min(\sigma^*, \sigma^+)$ and $\mu(x) = \max(\mu^-, \mu^*)$.

Modern Portfolio Theory

1.1.1.1 Introducing the quadratic utility function

The key idea of Markowitz (1956) was to transform the original non-linear optimization problem (1.1) into a quadratic optimization problem which is easier to solve numerically:

$$\begin{aligned} x^*(\phi) &= \arg \max x^\top \mu - \frac{\phi}{2} x^\top \Sigma x \\ \text{u.c. } & \mathbf{1}^\top x = 1 \end{aligned} \tag{1.3}$$

Modern Portfolio Theory

We can interpret ϕ as a risk-aversion parameter. If $\phi = 0$, the optimized portfolio is the one that maximizes the expected return and we have $\mu(x^*(0)) = \mu^+$. If $\phi = \infty$, the optimization problem becomes:

$$\begin{aligned} x^*(\infty) &= \arg \min \frac{1}{2} x^\top \Sigma x \\ \text{u.c. } & \mathbf{1}^\top x = 1 \end{aligned}$$

The optimized portfolio is the one that minimizes the volatility and we have $\sigma(x^*(\infty)) = \sigma^-$. It is called the minimum variance (or MV) portfolio.

Modern Portfolio Theory

Remark 1 We note that the previous problem can also be written as follows:

$$\begin{aligned} x^*(\gamma) &= \arg \min \frac{1}{2} x^\top \Sigma x - \gamma x^\top \mu \\ \text{u.c. } & \mathbf{1}^\top x = 1 \end{aligned} \quad (1.4)$$

with $\gamma = \phi^{-1}$. From a numerical point of view, this formulation has the advantage to be a standard quadratic programming (QP) problem (see Appendix A.1.1 on page 301). In this case, the minimum variance portfolio corresponds to $\gamma = 0$. Depending on the objective, we will use either the ϕ -problem or the γ -problem to calculate optimized portfolios.

Example 2

Minimum variance portfolio
via quadratic optimization



Modern Portfolio Theory

We consider Example 1. We have reported² in Table 1.1 the optimal portfolio for different values of ϕ . We verify that $\mu(x^*(\phi))$ and $\sigma(x^*(\phi))$ are two decreasing functions with respect to the parameter ϕ . It implies that the expected return $\mu(x^*)$ is an increasing function of the volatility $\sigma(x^*)$.

TABLE 1.1: Solving the ϕ -problem

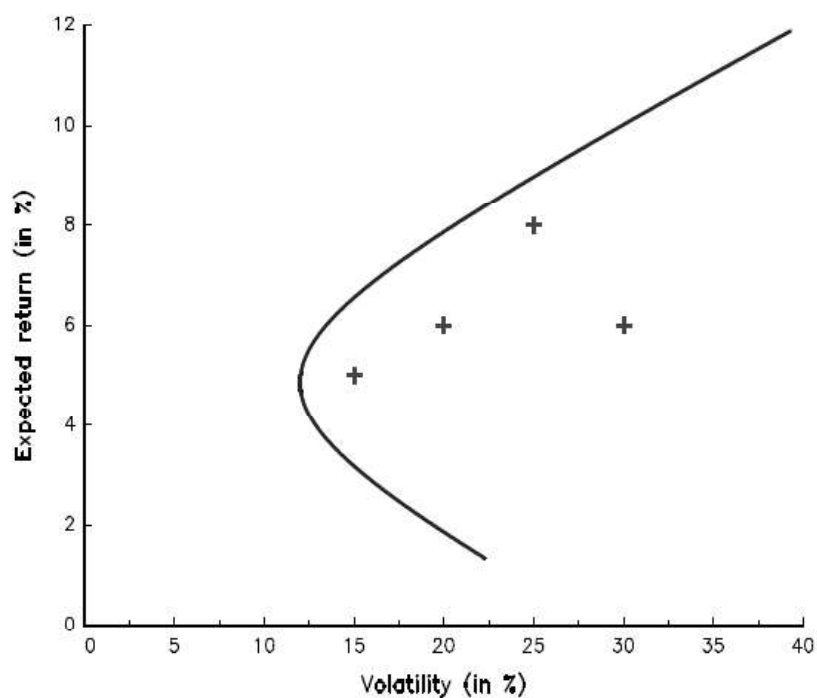
ϕ	$+\infty$	5.00	2.00	1.00	0.50	0.20
x_1^*	72.74	68.48	62.09	51.44	30.15	-33.75
x_2^*	49.46	35.35	14.17	-21.13	-91.72	-303.49
x_3^*	-20.45	12.61	62.21	144.88	310.22	806.22
x_4^*	-1.75	-16.44	-38.48	-75.20	-148.65	-368.99
$\mu(x^*)$	4.86	5.57	6.62	8.38	11.90	22.46
$\sigma(x^*)$	12.00	12.57	15.23	22.27	39.39	94.57

Modern Portfolio Theory

The formulation (1.3) allows to give a new characterization of the efficient frontier. It is the parametric function $(\sigma(x^*(\phi)), \mu(x^*(\phi)))$ with $\phi \in \mathbb{R}_+$. If we consider the previous example, we obtain the efficient frontier in Figure 1.2. We note that optimized portfolios substantially improve the risk/return profile with respect to the four assets, which are represented by a cross symbol.

Solving the μ -problem or the σ -problem is equivalent to finding the optimal value of ϕ such that $\mu(x^*(\phi)) = \mu^*$ or $\sigma(x^*(\phi)) = \sigma^*$. We know that the functions $\mu(x^*(\phi))$ and $\sigma(x^*(\phi))$ are decreasing with respect to ϕ and are bounded. The optimal value of ϕ can then be easily computed using the Newton-Raphson algorithm. We have reported some numerical solutions in Tables 1.2 and 1.3. For example, if μ^* is set to 7%, we obtain a portfolio with a volatility $\sigma(x^*)$ equal to 16.54%. It corresponds to portfolio *D* in Figure 1.1. If we target a volatility equal to 30%, the expected return of the optimized portfolio is 10.02% and the solution is portfolio *A* in Figure 1.1.

Modern Portfolio Theory



Modern Portfolio Theory

TABLE 1.2: Solving the unconstrained μ -problem

μ^*	5.00	6.00	7.00	8.00	9.00
x_1^*	71.92	65.87	59.81	53.76	47.71
x_2^*	46.73	26.67	6.62	-13.44	-33.50
x_3^*	-14.04	32.93	79.91	126.88	173.86
x_4^*	-4.60	-25.47	-46.34	-67.20	-88.07
$\sigma(x^*)$	12.02	13.44	16.54	20.58	25.10
ϕ	25.79	3.10	1.65	1.12	0.85

TABLE 1.3: Solving the unconstrained σ -problem

σ^*	15.00	20.00	25.00	30.00	35.00
x_1^*	62.52	54.57	47.84	41.53	35.42
x_2^*	15.58	-10.75	-33.07	-54.00	-74.25
x_3^*	58.92	120.58	172.85	221.88	269.31
x_4^*	-37.01	-64.41	-87.62	-109.40	-130.48
$\mu(x^*)$	6.55	7.87	8.98	10.02	11.03
ϕ	2.08	1.17	0.86	0.68	0.57

Modern Portfolio Theory

1.1.1.2 Adding some constraints

The introduction of constraints consists in modifying the specification of the optimization problem (1.3):

$$\begin{aligned} x^*(\phi) &= \arg \max x^\top \mu - \frac{\phi}{2} x^\top \Sigma x \\ \text{u.c.} \quad &\begin{cases} 1^\top x = 1 \\ x \in \Omega \end{cases} \end{aligned} \quad (1.5)$$

where $x \in \Omega$ corresponds to the set of restrictions³. These restrictions may be linear or non-linear. In the latter case, the optimization problem cannot be solved by the standard quadratic programming algorithm, but by enhanced non-linear optimization algorithms. The imposition of constraints will impact the set of optimized portfolios by reducing opportunity arbitrages. It implies that the constrained efficient frontier is located at the right of the unconstrained efficient frontier in the mean-variance map.

Modern Portfolio Theory

The most frequent constraint is certainly the no short-selling restriction. In this case, $x_i \geq 0$ and $\Omega = [0, 1]^n$. Let us define the leverage measure of the portfolio x as the sum of the absolute values of the weights:

$$\mathcal{L}(x) = \sum_{i=1}^n |x_i|$$

With the no short-selling restriction, the leverage measure is 100% because $\mathcal{L}(x) = \sum_{i=1}^n x_i = 1$ whereas it is larger than 100% without this constraint⁴.



Examples 3,4,5
Minimum variance portfolio
Constrained optimization

Modern Portfolio Theory

Let us introduce some constraints in Example 1. In Figure 1.3, we have reported two constrained efficient frontiers, the first one by imposing no short-selling and the second one by imposing that the weights are between 0% and 40%. We verify that we may substantially reduce opportunity arbitrages. Solutions of the σ -problem are given in Table 1.4. If we target a volatility equal of 15%, the expected return of the optimized portfolio is 6.55% for the unconstrained problem, 6.14% for the shortsale constrained problem and 6.11% if we impose an upper bound of 40%. So, by imposing no short positions, we have reduced the expected return by 41 bps. The impact of the upper bound is small. If the target volatility becomes 20%, the results become 7.87%, 7.15% and 6.74% and the impact is larger than in the previous case.

Modern Portfolio Theory

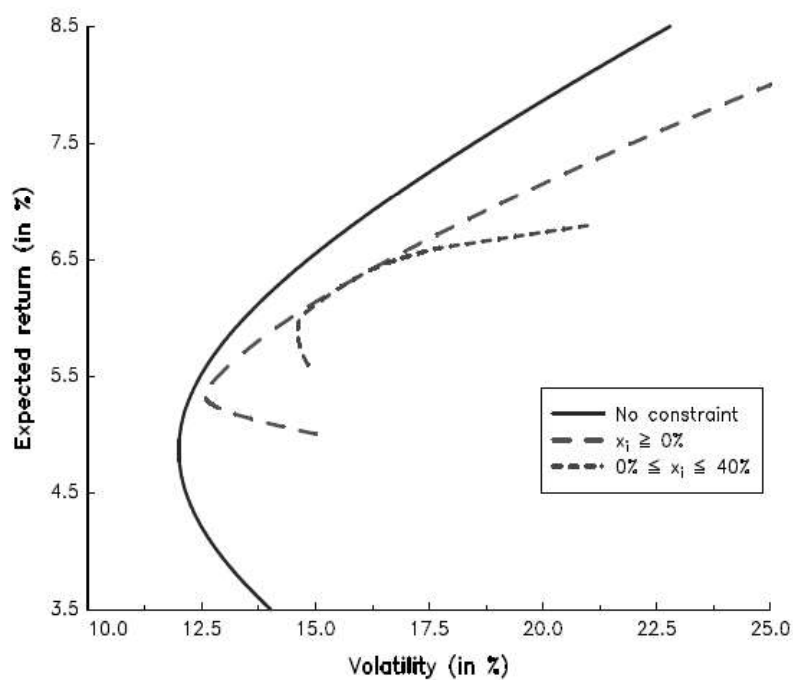


FIGURE 1.3: The efficient frontier with some weight constraints

Modern Portfolio Theory

TABLE 1.4: Solving the σ -problem with weight constraints

σ^*	$x_i \in \mathbb{R}$		$x_i \geq 0$		$0 \leq x_i \leq 40\%$	
	15.00	20.00	15.00	20.00	15.00	20.00
x_1^*	62.52	54.57	45.59	24.88	40.00	6.13
x_2^*	15.58	-10.75	24.74	4.96	34.36	40.00
x_3^*	58.92	120.58	29.67	70.15	25.64	40.00
x_4^*	-37.01	-64.41	0.00	0.00	0.00	13.87
$\mu(x^*)$	6.55	7.87	6.14	7.15	6.11	6.74
ϕ	2.08	1.17	1.61	0.91	1.97	0.28

Modern Portfolio Theory

1.1.1.3 Analytical solution

The Lagrange function of the optimization problem (1.3) is:

$$\mathcal{L}(x; \lambda_0) = x^\top \mu - \frac{\phi}{2} x^\top \Sigma x + \lambda_0 (1^\top x - 1)$$

where λ_0 is the Lagrange coefficients associated with the constraint $1^\top x = 1$. The solution x^* verifies the following first-order conditions:

$$\begin{cases} \partial_x \mathcal{L}(x; \lambda_0) = \mu - \phi \Sigma x + \lambda_0 1 = 0 \\ \partial_{\lambda_0} \mathcal{L}(x; \lambda_0) = 1^\top x - 1 = 0 \end{cases}$$

We obtain $x = \phi^{-1} \Sigma^{-1} (\mu + \lambda_0 1)$. Because $1^\top x - 1 = 0$, we have $1^\top \phi^{-1} \Sigma^{-1} \mu + \lambda_0 (1^\top \phi^{-1} \Sigma^{-1} 1) = 1$. It follows that:

$$\lambda_0 = \frac{1 - 1^\top \phi^{-1} \Sigma^{-1} \mu}{1^\top \phi^{-1} \Sigma^{-1} 1}$$

The solution is then⁵:

$$x^*(\phi) = \frac{\Sigma^{-1} 1}{1^\top \Sigma^{-1} 1} + \frac{1}{\phi} \cdot \frac{(1^\top \Sigma^{-1} 1) \Sigma^{-1} \mu - (1^\top \Sigma^{-1} \mu) \Sigma^{-1} 1}{1^\top \Sigma^{-1} 1}$$

 (1.6)

Modern Portfolio Theory

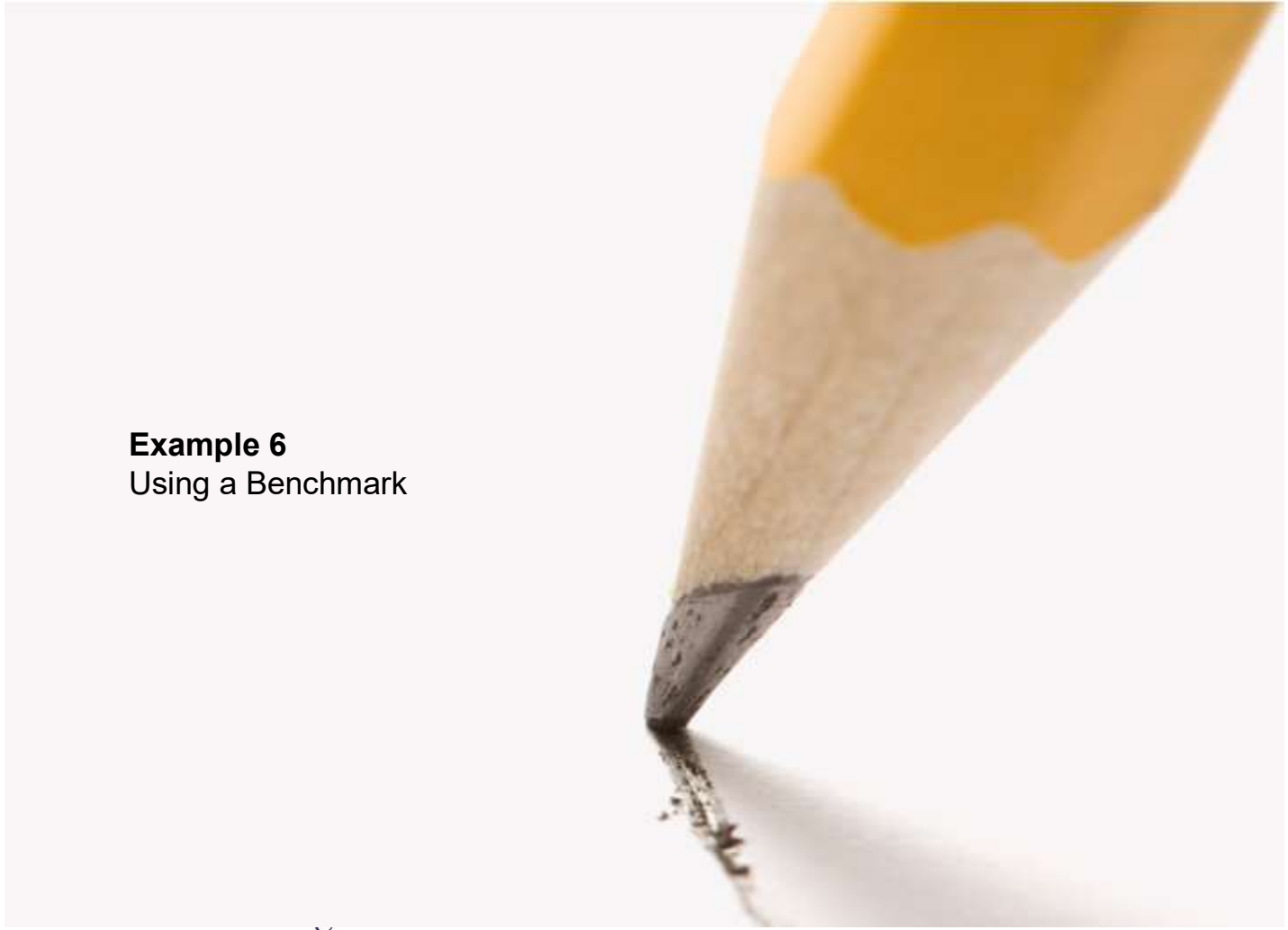
We deduce also that the global minimum variance portfolio has the following expression:

$$x_{\text{mv}} = x^*(\infty) = \frac{\Sigma^{-1} 1}{1^\top \Sigma^{-1} 1}$$

If we introduce other constraints, it is not possible to obtain a comprehensive analytical solution. Let us consider for example the no short-selling constraint. The Lagrange function becomes:

$$\mathcal{L}(x; \lambda_0, \lambda) = x^\top \mu - \frac{\phi}{2} x^\top \Sigma x + \lambda_0 (1^\top x - 1) + \lambda^\top x$$

where $\lambda = (\lambda_1, \dots, \lambda_n)$ is the vector of Lagrange coefficients associated with the constraints $x_i \geq 0$. The first-order condition is then $\mu - \phi \Sigma x + \lambda_0 1 + \lambda = 0$. It follows that $x = \phi^{-1} \Sigma^{-1} (\mu + \lambda_0 1 + \lambda)$. The Kuhn-Tucker conditions are $\min(\lambda_i, x_i) = 0$ for all $i = 1, \dots, n$. It implies that if $x_i = 0$ then $\lambda_i > 0$ and if $x_i > 0$ then $\lambda_i = 0$. We find also a formula close to the previous one, but the universe is limited to assets which present positive weights. This formula is therefore endogenous.



Example 6

Using a Benchmark

Modern Portfolio Theory

1.1.4 Portfolio optimization in the presence of a benchmark

Let us now consider a benchmark which is represented by a portfolio b . The tracking error between the active portfolio x and its benchmark b is the difference between the return of the portfolio and the return of the benchmark:

$$\begin{aligned} e &= R(x) - R(b) \\ &= \sum_{i=1}^n x_i R_i - \sum_{i=1}^n b_i R_i \\ &= x^\top R - b^\top R \\ &= (x - b)^\top R \end{aligned}$$

The expected excess return is:

$$\mu(x | b) = \mathbb{E}[e] = (x - b)^\top \mu$$

whereas the volatility of the tracking error is¹¹:

$$\sigma(x | b) = \sigma(e) = \sqrt{(x - b)^\top \Sigma (x - b)}$$

Modern Portfolio Theory

The objective of the investor is to maximize the expected tracking error with a constraint on the tracking error volatility:

$$\begin{aligned} x^* &= \arg \max (x - b)^\top \mu \\ \text{u.c. } &1^\top x = 1 \quad \text{and} \quad \sigma(e) \leq \sigma^* \end{aligned} \quad (1.9)$$

Like the Markowitz problem, we transform this σ -problem into a ϕ -problem:

$$x^*(\phi) = \arg \max f(x | b)$$

with:

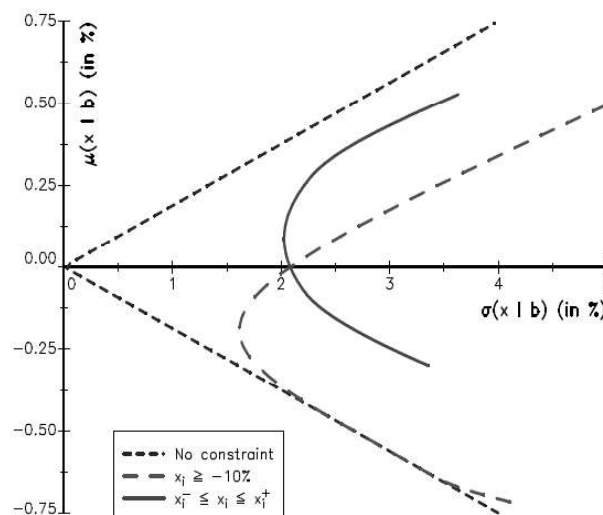
$$\begin{aligned} f(x | b) &= (x - b)^\top \mu - \frac{\phi}{2} (x - b)^\top \Sigma (x - b) \\ &= x^\top (\mu + \phi \Sigma b) - \frac{\phi}{2} x^\top \Sigma x - \left(\frac{\phi}{2} b^\top \Sigma b + b^\top \mu \right) \\ &= x^\top (\mu + \phi \Sigma b) - \frac{\phi}{2} x^\top \Sigma x + c \end{aligned}$$

where c is a constant which does not depend on the portfolio x . We recognize a quadratic program which could be solved easily using numerical algorithms. The efficient frontier is then the parametric curve $(\sigma(x^*(\phi) | b), \mu(x^*(\phi) | b))$ with $\phi \in \mathbb{R}_+$.

Modern Portfolio Theory

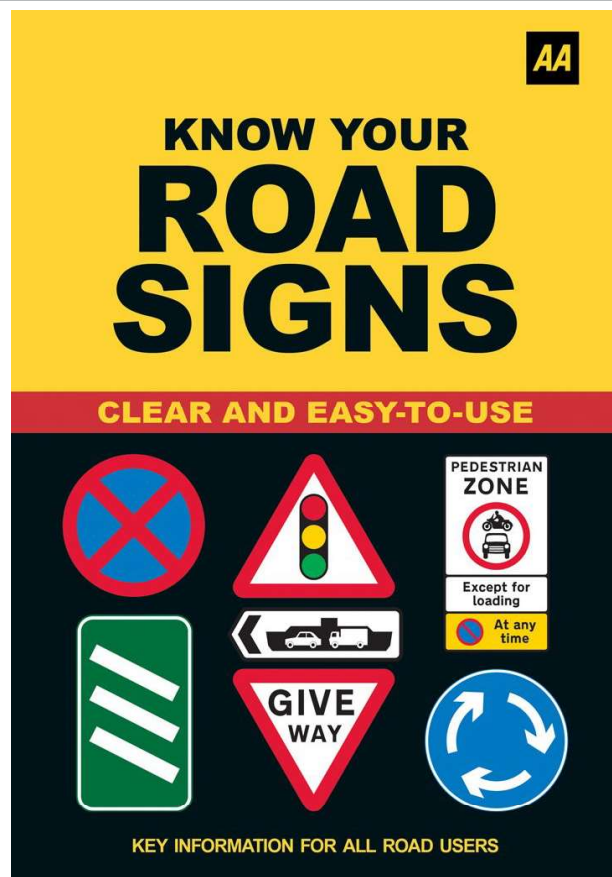
We consider Example 1 with the benchmark $b = (60\%, 40\%, 20\%, -20\%)$.

In Figure 1.6, we have represented the corresponding efficient frontier. We verify that it is a straight line when there is no restriction (Roll, 1992). If we impose that $x_i \geq -10\%$, the efficient frontier is moved to the right. For the third case, we assume that the weights are between a lower bound and an upper bound: $x_i^- \leq x_i \leq x_i^+$ with $x_i^+ = 50\%$. For the first three assets, the lower bound x_i^- is set to 0, whereas it is equal to -20% for the fourth asset.



MVO: Drawbacks and Consequences

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MVO: Drawbacks and Consequences

In reality, these model parameters are not observable ex-ante, thus they must be estimated. This task is commonly accomplished with the download of time series of **historical returns for the selected asset classes** and then consequently with the statistical computation of the relevant parameters using past data.

Extracts from Maria Debora Braga, Risk-Based Approaches to Asset Allocation: Concepts and Practical Applications, Springer 2016

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Knowledge | Skills | Conduct

MVO: Drawbacks and Consequences

The standard implementation of the Markowitz model then usually follows an approach named by Kan and Zhou (2007) the 'plug-in' rule. In practice, the **estimated parameters are processed into the mean-variance optimiser that treats them as if they were the true parameters**. Handling the asset allocation problem in this way has an important and serious implication: The procedure becomes **totally deterministic as the parameters uncertainty is completely neglected**.

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MVO: Drawbacks and Consequences

Of course, we are not able to know a priori how far we are from the true parameters. As indicated by a wide number of authors, Fernandes et al. (2012), Drobetz (2001), Eichorn et al. (1998), Grauer and Shen (2000), Herold and Maurer (2006), Jobson and Korkie (1981), Jorion (1985, 1992), Michaud (1989), implementing the Mean-Variance Analysis **without recognising the existence of estimation risk inherent in the input parameters** has a huge impact on the optimised portfolios and leads to several undesirable features and deficiencies.

Drawback 1



Drawback 1

[1] The issue deserving attention is the **lack of diversification of optimal portfolios** which paradoxically contrasts with one of the main ideas of Markowitz. Efficient allocations frequently completely exclude some asset classes of the chosen investment universe and give extremely large weights to some others. They also show sudden shifts along the Efficient Frontier. In addition, not so rarely the most northeast point of the Mean-Variance Efficient Frontier lacks completely diversification and is equivalent to a mono-asset portfolio.³ These are consequences of the high discriminatory and impassive manner adopted by the portfolio construction method in dealing with data inputs which are considered extremely accurate. The mean-variance optimiser tends to heavily weigh those asset classes that show high estimated returns compared to low variances (or standard deviations) and negative correlations. These are also the asset classes that are most likely to suffer from large estimation errors. For this reason, Michaud (1989) wrote: “The unintuitive character of many optimized portfolios can be traced to the fact that MV optimizers are, in a fundamental sense, estimation-error maximizers.”

Drawback 2



Drawback 2

[2] The second major drawback of mean-variance efficient portfolios is their **instability**. This attribute defines the high sensitivity of portfolio allocations to small changes in the estimated parameters. Especially, in the case we have in the investment universe couples of asset classes with similar risk-return profile, small perturbations may completely alter the distribution of the portfolio weights because they may cause an alternation between the dominant and the dominated asset class. This circumstance proves that optimised portfolios are not reasonable under different set of plausible input parameters. Since they are estimated from historical observation of asset classes' returns, the occurrence of perturbations must be taken for granted and therefore the problem of instability has a practical impact. The literature (Best and Grauer 1991) has suggested that especially changes in expected returns tend to impact dramatically on optimal portfolio weights.

Drawback 3



Drawback 3

[3] The third limitation consists of failing to recognise the **non-uniqueness of optimal portfolios**. As sharply noted by Michaud (1989):

“Optimizers, in general, produce, a unique optimal portfolio for a given level of risk. This appearance of exactness is highly misleading, however. The uniqueness of the solution depends on the erroneous assumption that the inputs are without statistical estimation error.”

Hence, in practice, it would be helpful, in order to reach the goal of identifying recommended portfolio structures, to consider many statistically equivalent portfolios and take the average weights resulting from their different compositions.

Drawback 4



Drawback 4

[4] The last but potentially the most serious drawback affecting mean-variance efficient portfolios is the **poor out-of-sample performance**. Outside the sample period used to estimate input parameters, classical Markowitz's portfolios show a considerable deterioration of performance with respect to the expected one and the same occurs in terms of risk-adjusted performance.

As observed by DeMiguel et al. (2009) and Jobson and Korkie (1981), the extent to which they fall short of the original targeted performance is such that very simple investment criteria can dominate Mean-Variance Optimisation. Thus, this means that the 'plug-in rule' cannot be validated.

Consequences

Ignoring the possible misspecification of the input parameters has major consequences especially if we refer to expected returns. Sample means are subject to large fluctuations and their predictive power seems poor. The errors in variances and covariances/correlations are considered less costly than estimation errors in the expected returns.

Chopra and Ziemba (1993) found that errors in mean are at least 10 times as important as errors in variances and errors in variances are about twice as important as errors in covariances.

Consequences

Michaud and Michaud (2008) contributed to this argument writing:

“The procedure **overuses statistically estimated** information and magnifies the impact of estimation errors. It is not simply a matter of garbage in, garbage out, but rather a molehill of garbage in, a mountain of garbage out.”

Consequences

Given the implications of the presence of estimation risk for investment portfolios, several attempts have been made to overcome them. From the mid 1980s onwards, alternative methodologies have been developed to deal with the problem of parameters uncertainty. They can be distinguished between **heuristic** and **Bayesian** approaches.

Consequences

Heuristic approaches

The first group includes using the **resampling** technique patented by Michaud and Michaud (2008) and the application of additional weight constraints (besides the classical long-only and budget constraints) suggested by Frost and Savarino (1988), and Jagannathan and Ma (2003).

Consequences

Bayesian approaches

Can be implemented in different ways. One application consists in the use of shrinkage estimators that propose estimation of expected returns by shrinking the sample mean toward a common mean (Jorion 1985). A second application is represented by the **Black and Litterman** model (1992) which combines two sources of information (equilibrium returns and investor's views) to obtain predictive returns.

Consequences

In practice, heuristic approaches and Bayesian approaches **take different actions in response to the impact of the unavoidable imprecision of estimated inputs**, especially of expected returns, on optimal portfolios construction. The former acts on the model side to make the optimisation process less deterministic or to force it to produce more diversified portfolios. The latter acts on the estimation side to improve expected returns estimates with respect to their sample estimates. Obviously, it also makes sense to use methodologies that combine features of heuristic and Bayesian approaches as proposed by Fernandes et al. (2012).

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